

EAN-Infrared Temperature

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 **CAUTION:** Alerts to a condition or improper procedure that can result in equipment or component damage if not avoided.

 **IMPORTANT:** Identifies information that is critical to successful setup, configuration, or operation of the system.

 *Provides supplementary information that aids in the use or understanding of the system or subject matter but is not critical to operation.*



1 Overview

This document describes features surrounding infrared (IR) camera temperature measurements. Sightline hardware products can be used to acquire and process 14- or 16-bit data from IR cameras. This allows calculation of statistics on the raw pixel count data, which can be used to provide metrics such as the minimum, maximum, and average pixel value. The auto gain algorithm is configurable, allowing clipping of raw data to a specific digital range and the corresponding temperatures. Standard or custom false color lookup tables (LUT) can then be applied to highlight temperature region.

Sightline products allow snapshot exports of raw camera data. Raw can be saved in 16-bit PNG files. The process for saving these snapshots to the microSD card as well as extracting the data is detailed in [EAN-File-Recording](#).

1.1 Camera Specific Features

The FLIR Tau 2, Boson, and DRS Tamarisk Superframe cameras are all compatible with Sightline hardware.

FLIR Tau 2 cameras can be purchased to be factory calibrated in *TLinear* mode. This is not available on a non-calibrated Tau 2. The FLIR Boson+ can be purchased as a *radiometric* version.

The DRS Tamarisk Superframe camera also supports raw temperature calibrated data output. This data is full 16-bits.

The calibrated cameras allow the output data (pixel counts) to be converted to temperature using linear conversion provided by the manufacturer.

 See the camera calibration overview in the FLIR Advanced Radiometry App Note document available at [FLIR](#) before attempting to use this feature.

1.2 Additional Support Documentation

The [Panel Plus](#) User Guide is accessible from the Help menu within the application.

The Interface Command and Control (IDD) describes the native communications protocol.

Additional Engineering Application Notes (EANs) and ICDs are available on the [Product Support Documentation](#) page in the [Sightline Knowledge Base](#).

1.3 Sightline Software Requirements

 **IMPORTANT:** The Panel Plus software version should match the firmware version running on the OEM. Firmware and Panel Plus software versions are available on the [Software Releases](#) page.

Table 1: Sightline OEM Software Requirements

OEMs	Initial Software Release	Provisions
4110-OEM	3.11.x	--
1710-OEM	3.9.x	--
4100-OEM	3.7.x	Three channel processing requires 3.9.x and above.
1750-OEM	3.6.x	Dual channel processing and Snap All Cams requires 3.7.x and above.
4000-OEM	3.00.xx	AI-Assist and AI-Detect are not supported. Blending requires 3.3.x and above.
Legacy OEMs are supported up to 3.5.x		
1500-OEM	2.20.xx	--
3000-OEM	2.22.xx	Rev B6 requires updated firmware to accommodate the change in memory.



1.4 Software Feature Version Requirements

Software feature availability is determined by software version. Refer to the table below for the minimum software version that supports each feature described in this document. For questions or upgrades, contact [Sales](#).

Table 2: Software Feature Version Requirements

Software Function	Initial Software Release	Notes
Enhancement, High Depth and Radiometric Temp	2.22.xx	Snapshot recording requires Recording
High Bit Depth/Temp Telemetry	3.00.xx	Optional: Enhancement and Recording Snapshot recording requires Recording

2 Requirements

The infrared camera should support the following:

- Output of digital pixel data with full resolution (14 or 16 bits).
- Operating mode where output digital counts are linear with temperature. Normal detector counts are linear with photons, i.e., flux, which is not the same as temperature.

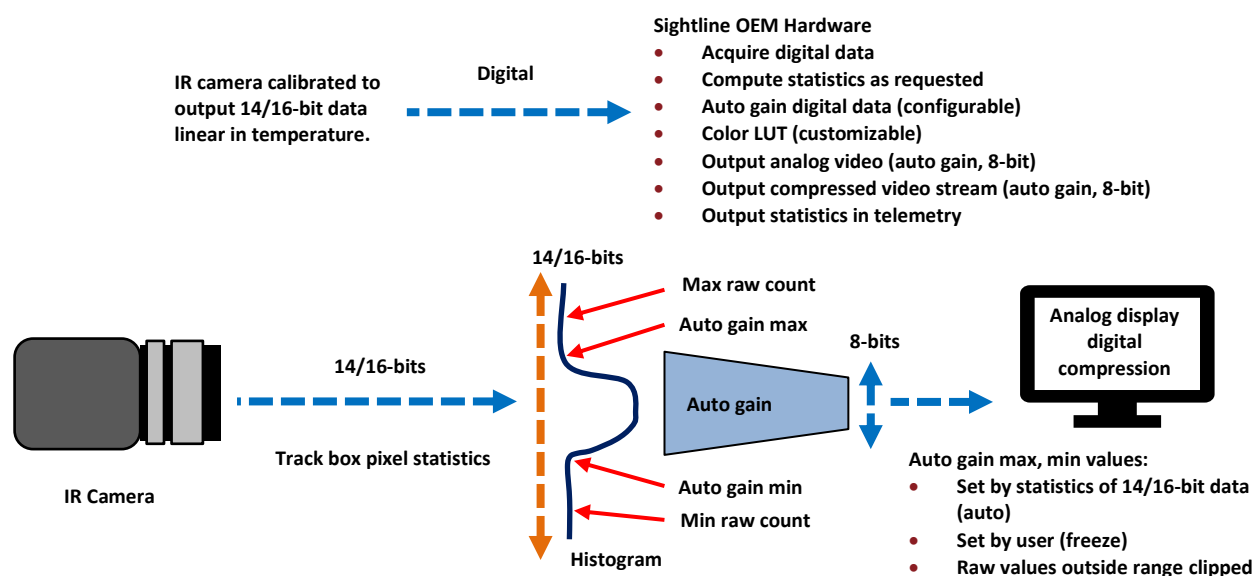


Figure 1: IR Temperature Measurement System

3 Automatic and Manual Gain

Gain control, also known as auto gain, is used to convert a wide range of digital counts to an 8-bit range for display or compression. Gain control can be automatic (auto) or manual.

Auto gain maps the minimum and maximum digital counts in the incoming 16-bit image to 0 and 255, respectively in the resulting 8-bit image. The auto gain range can be locked to a specific region using the SVP command [SetDigitalCameraParameters \(0x70\)](#). Auto gain can also be set up in the Panel Plus *Capture* tab. *Brightness* and *Contrast* can be used to shift and stretch the mapping.



3.1 Auto Gain Region of Interest

The area over which the statistics are calculated can be set as either a percentage of the whole image or a preset quadrant within the image.

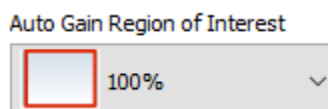


Figure 2: Auto Gain Region Configuration

3.2 Automatically Adjust Gain

To automatically adjust gain on a high-bit depth camera, select *High Bit Depth Auto Gain*. For each frame, the system will calculate statistics used to adjust the image to an 8-bit output. The auto gain calculation currently uses a spread of six standard deviations and uses the mean as the center of the range, i.e., +/- 3.0 standard deviations about the mean. The *Manual Min* and *Manual Max* slider values define the range of values that get linearly mapped to 0 and 255, respectively.

Request: When this button is clicked, Panel Plus queries the OEM board for the current image control values (including exposure and gains values) and then fills them in. These values are then displayed on the *Manual Min* and *Manual Max* sliders.

The input video is 16-bit images (gray values 0 to 65535). The internal (after auto gain calculation) and output video is 8-bit images (gray values 0 to 255). In this example, a value of 18870 in the input image will map to 0 in the internal/output and a value of 20271 will map to 255.

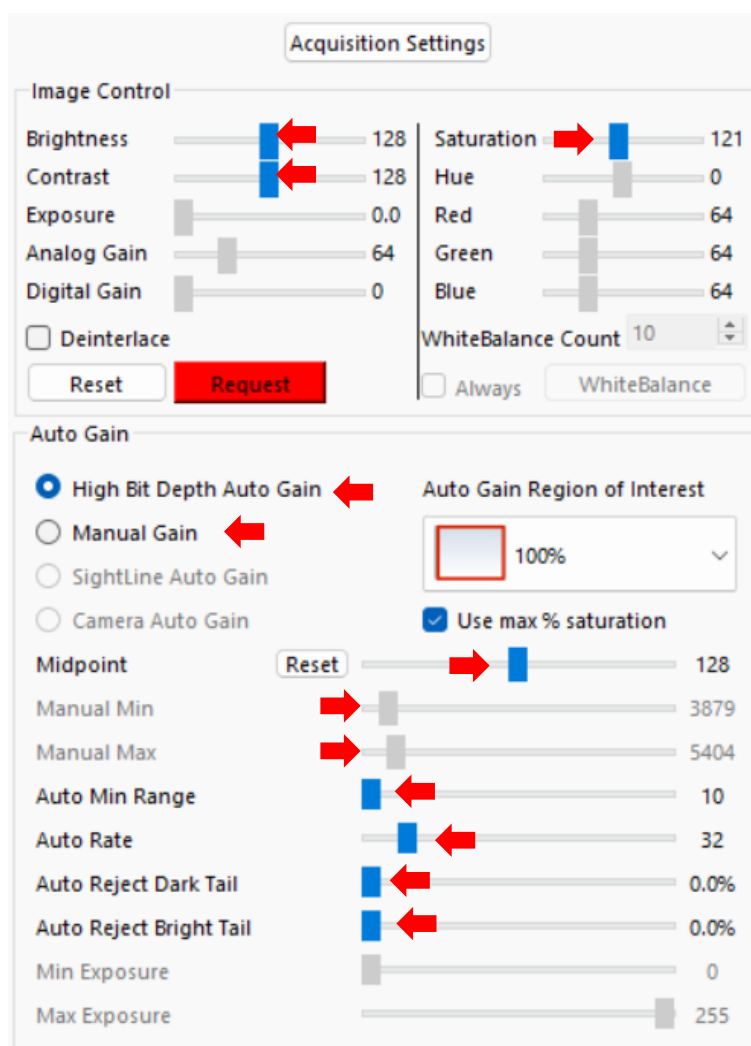


Figure 3: High Bit Depth Auto Gain

The *Midpoint* slider sets the 16-bit value that corresponds to the central 8-bit value of 128 within the defined minimum and maximum range. This effectively means that approximately half of the pixels will be mapped to values below this midpoint, and half will be mapped to values above it.



Brightness: Shifts the mapping to make the resulting image brighter or darker.

Contrast: Stretches the mapping to make the 8-bit image higher or lower contrast.

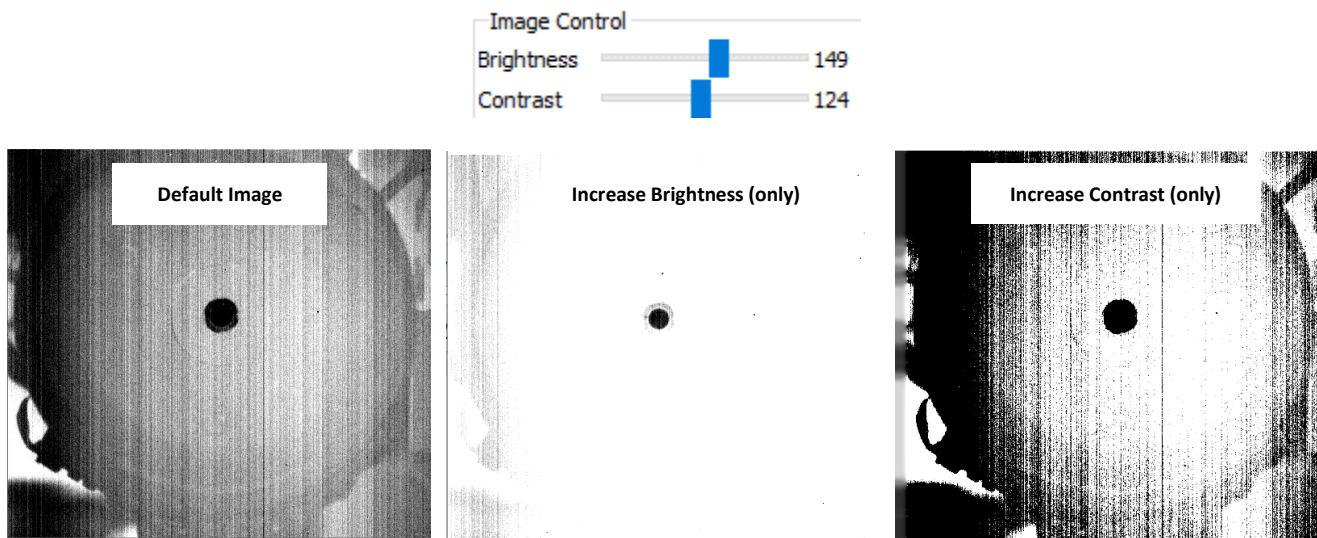


Figure 4: Brightness and Contrast

Auto Min Range: Sets the minimum spread (max pixel value - min pixel value) that the algorithm will produce data. This prevents over-scaling of flat scenes.

Auto Rate: Controls the rate (frames) that the auto gain algorithm adjusts: 1 = minimum rate, 255 = maximum rate, 32 = default.

New in 3.4.x software:

Auto Reject Dark Tail: Reject or ignore a percentage of the left/dark side of the histogram when calculating autogain. 0 = reject none, 10 = reject 1.0%, 255 = reject 25.5%. This is useful when there are a small number of very dark pixels in a scene.

Auto Reject Bright Tail: Reject or ignore a percentage of the right/bright side of the histogram when calculating autogain. 0 = reject none, 10 = reject 1.0%, 255 = reject 25.5%. This is useful when there are a small number of very bright pixels in a scene.

New in 3.9.x software:

Use % max Saturation checkbox: Enables the saturation control, which selects min and max based on the desired % of saturated and undersaturated pixels.

Saturation: Controls the maximum percentage of pixels that can be fully saturated (value 255) or undersaturated (value 0) in the resulting image after 16-bit to 8-bit conversion. The maximum percentage of excluded pixels is calculated as: $\text{saturation} / (256 \times 10)$.

For more information [See Sensor Controls and Gain Modes](#) in the IDD.



3.3 Manually Adjust Gain

To manually adjust gain on a high-bit depth camera and freeze the auto gain, set the *Auto Gain* mode to *Manual Gain*. Control the gain range with the *Manual Min* and *Manual Max* sliders.

Use the *Midpoint* slider to control the quality of darker and lighter input pixels. In *Manual Gain* mode, brightness and contrast are also applied.

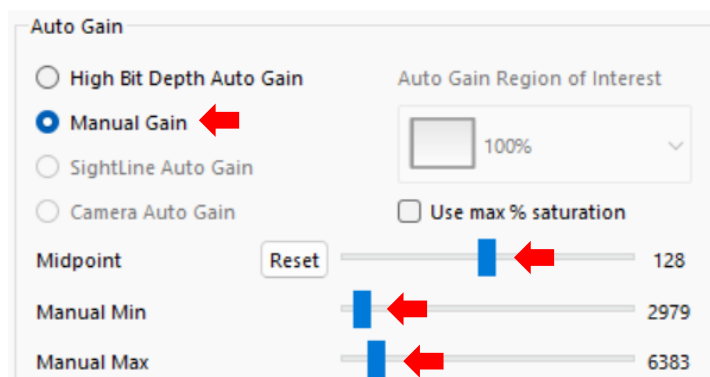


Figure 5: Manually Adjust Gain

3.4 Converting Raw Counts to Temperature

Panel Plus has built-in functions to convert the raw pixel count to temperature for the FLIR Tau 2, Boson, and DRS Superframe cameras.

In the *Thermal* section of the *Capture* tab the radio buttons select the display units.

The drop-down menu selects the conversion factor for the attached camera.

A custom linear conversion function is also provided, allowing users to translate raw counts into temperature values using the formula $Out = raw * A + B$, where the user can specify A and B.

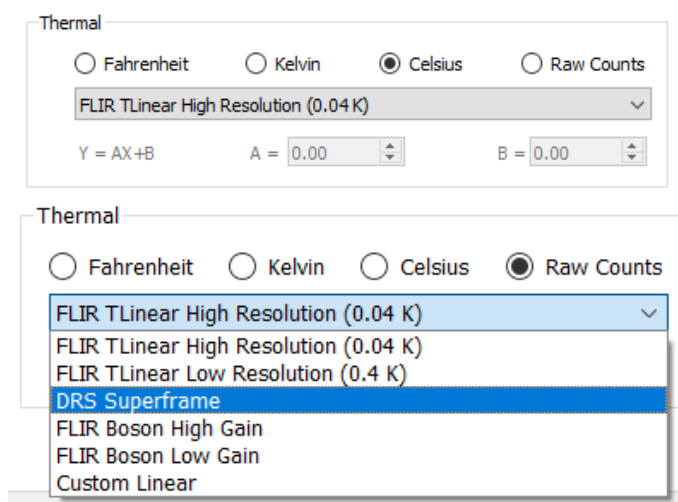


Figure 6: Temperature Scale and Conversion Factor

4 Radiometric Boson

The radiometry-capable Boson model is supported in software 3.7.x and above.

4.1 Configuration through FLIR GUI

To get accurate temperature readings, settings on the camera must be configured and persisted through the FLIR GUI.

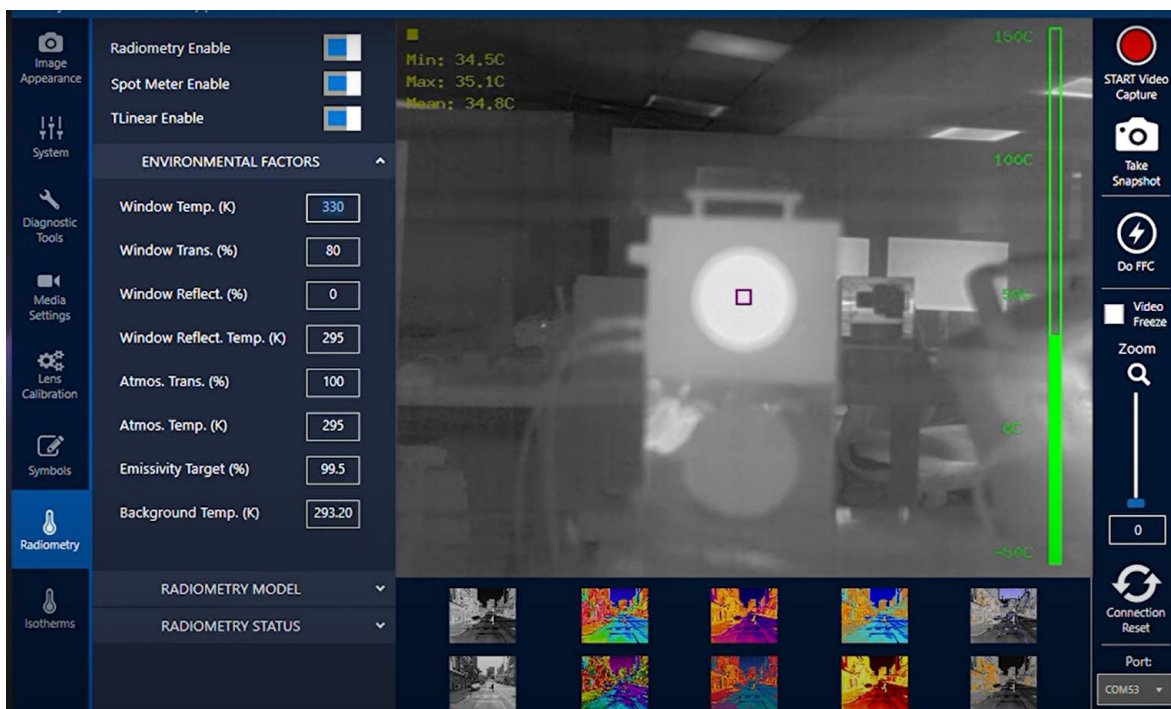


Figure 7: Boson+ Application v4.2

For more information on configuration through the FLIR GUI while the camera is connected to the Sightline board, see the *Boson Cameras - TCP Passthrough Setup for FLIR GUI* and the *Boson Cameras: Enabling FLIR GUI Color Palette* sections in [EAN-FLIR-Cameras](#).

For more information on radiometry configuration see the [FLIR Video Guide](#).

4.2 Measuring Temperature in Panel Plus

See the [Measuring Temperature in Regions](#) and [Color LUT Display of User Defined Temperature Regions](#) section for more information on how to use the thermal temperature conversion functions from telemetry data in Panel Plus.

Temperature values for Boson High Gain and Boson Low Gain are supported:

- High Gain = raw count / 100
- Low Gain = raw count / 50

The Custom Linear option can also be used for a custom function:

- Custom linear = raw count * A + B

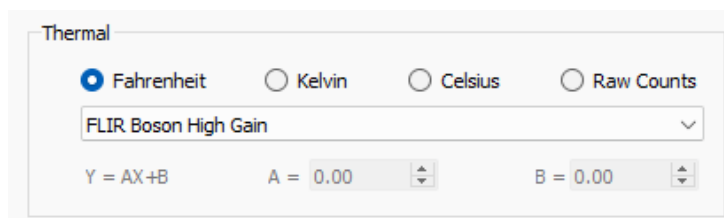


Figure 8: Thermal Temperature from Raw Count Conversion for Boson High Gain



5 FLIR Tau 2 T Linear Camera Setup

5.1 Acquiring Tau 14-Bit Data

Open the Panel Plus application and connect to the 1500-OEM. See the [EAN-Digital-Video-Configuration](#) and the [camera configuration table](#) for digital camera setup information.

Digital camera setting: FLIR Tau 2 640 14-bit (or FLIR Tau 2 336 14-bit).

5.2 FLIR T Linear Modes and Ranges

- FLIR TLinear supports low and high-resolution modes.
- 14-bit digital output maximum counts = 16383.
- Low resolution digital counts correspond to 0.4 Kelvin.
- High resolution digital counts correspond to 0.04 Kelvin.

The Tau 2 camera has a gain mode setting of High, Low and Auto that can be set using the FLIR Camera Controller GUI.

- High gain mode (default) limits the maximum temp to ~350°F.
- Low gain mode allows for measuring higher temperature scenes.
- Auto mode switches between the two modes at a programmable scene temperature. This is the preferred selection.

Table 3: FLIR TLinear High/Low Resolution Temperature Ranges

	Min °K	Max °K	Min °C	Max °C	Min °F	Max °F
Low Res (0.4)	0	6553	-273	6280	-459	11336
High Res (0.04)	0	655	-273	382	-459	719

5.3 FLIR Tau 2 TLinear Configuration

The FLIR Tau 2 camera must be configured in 14-bit TLinear mode. This can be done using the USB dongle provided with the camera, or through the serial passthrough feature.

To use serial passthrough, setup serial port 2 using 57,600 baud. See the [EAN-Ethernet-and-Serial-Communication](#) document for details.

1. From the FLIR *Camera Controller* interface go to the *Digital Video* tab.
2. Select all the settings shown in [Figure 9](#).

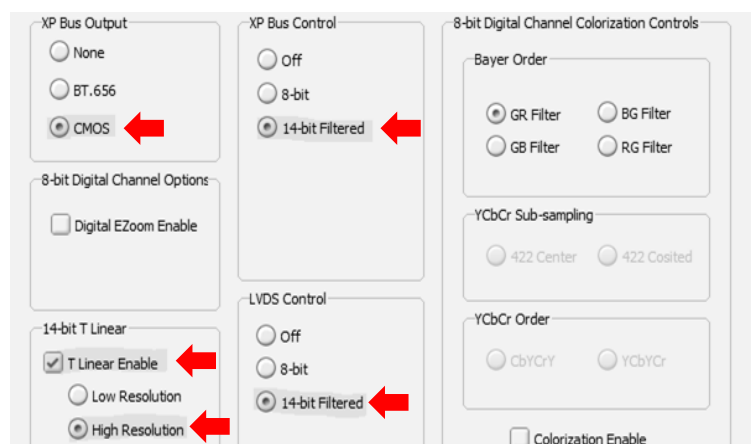


Figure 9: FLIR Camera Controller Interface - Digital Video Tab

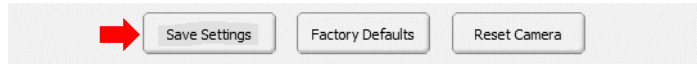
If the 14-bit TLinear options are not visible, the FLIR Tau 2 camera has not been factory calibrated.



- From the *Setup* tab, set *Gain Mode* to *Auto*.



- Click *Save Settings* to save the settings for subsequent restarts.



6 DRS Tamarisk Precision Superframe Setup

The DRS Tamarisk Superframe is supported in software 2.22.15 and above. This has currently been tested on the 320 version of this camera, the 640 is not supported.

See the [EAN-DRS-Cameras](#) for Panel Plus configuration settings.

6.1 DRS Superframe Modes and Ranges

Data output is 16-bits (there are no programmable ranges). Each 16-bit pixel value represents Kelvin temperature as 11.5 fixed point binary. For Celsius, use 16-bit pixel value as the integer, divide by 32.0f, then subtract 273.15.

7 Measuring Temperature in Regions

Tracking provides mean, min, and max raw digital counts of up to five Region of Interests (ROI) within the image. Region of Interests can be specified through the Sightline Command and Control Protocol (SVP) [ModifyTracking \(0x05\)](#) or by starting tracks with Panel Plus. More information on tracking parameters can be found in [EAN-Target-Tracking](#) under *Additional Tracking Parameters*.

- From the Panel Plus main menu » *Track* » *Manual* tab, select a manual target designation mode from the drop-down menu.

Other controls can be used to adjust the region that the statistics will be calculated.

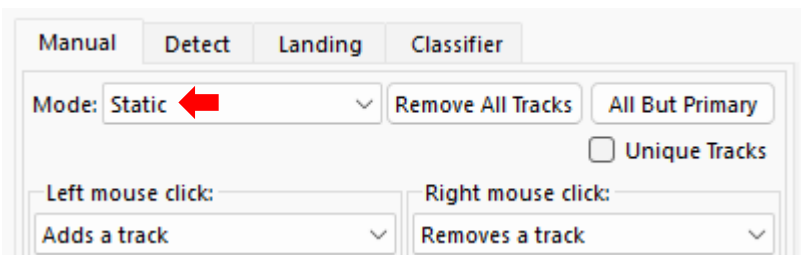


Figure 10: Manual Target Designation Modes

- Click *Show Telemetry* then enable *Track Box Pixel Statistics*. If a conversion factor has been selected (*Capture* tab), then the temperature values will be displayed instead of raw counts.

☒ Track Box Pixel Statistics:

Idx:	Mean:	Min:	Max:	
1	274.6	235.1	310.9	°C
2	276.7	236.7	311.0	°C
3	276.1	235.8	312.3	°C

- Click the desired region on the image in Panel Plus to start a track.



Statistics values are read through the SVP [TrackingBoxPixelStats \(0x78\)](#). Raw pixel count value is provided if the current camera is a high-bit depth digital imager, and the pixels stats output is enabled in the [CoordinateReportingMode \(0x0B\)](#).

Conversion from raw digital counts can be done based on the configuration of the IR imager.

8 Color LUT Display of User Defined Temperature Regions

The *Capture* tab also provides the ability to specify temperature ranges of interest. Freeze the auto gain algorithm to display the desired temperature range. Use the custom color LUT to highlight temperatures of interest. Additional information and options for setting the LUT can be found in the *False Color* section in [EAN-Enhancement](#).

8.1 Defined Temperature Range - Freeze Auto Gain

Example:

1. Set the range and freeze the auto gain:
 - a. Select *Manual Gain*.
 - b. The FLIR TLinear camera should be in high resolution mode (0.04 °K/count).
 - c. Set min temperature of interest to 80°F (300°K = 7500 digital counts).
 - d. Set max temperature of interest to 400°F (478°K = 11,950 digital counts).

Auto Gain

☐ High Bit Depth Auto Gain
☒ Manual Gain
☐ SightLine Auto Gain
☐ Camera Auto Gain

Auto Gain Region of Interest: 100% ▼

Midpoint: 128

Manual Min: 80.0°F

Manual Max: 400.0°F

Auto Min Range: 13.2°F

Auto Rate: 32

Auto Reject Dark Tail: 0.0%

Auto Reject Bright Tail: 0.0%

Min Exposure: 0

Max Exposure: 255

Thermal

☒ Fahrenheit
☐ Kelvin
☐ Celsius
☐ Raw Counts

FLIR TLinear High Resolution (0.04 K) ▼

Y = AX+B A = 0.00 B = 0.00

Figure 11: Defined Temperature Range



All temperatures below 80°F are now mapped to 0. All temperatures above 400°F are mapped to 255. Each output count in between now corresponds to a 0.8°F change in temperature. Temperature to count conversion is $(temp - 80) * (255) / (320)$.

2. Set custom LUT setting 0 » 255 counts maps to RGB values (r,g,b):
 - a. From the Panel Plus *Enhance* tab, in the *False Color* drop-down menu select *User Palette*.

False Color: User Palette

- b. From the Panel Plus main menu » *Configure* » *User Palette*.
- c. Initialize the LUT to an incrementing grayscale LUT[x] = (x,x,x).
- d. Map the range of human body temperature (80°F - 100°F) LUT[0 - 16] to a color range (e.g., green).
- e. Map the range of hot objects such as electrical wires (150°F - 200 °F) LUT[56 - 96] to a color range (e.g., yellow).
- f. Map all values above 400°F to pure red by setting LUT[255] to (255,0,0).
- g. Click *Send to Board* to send the settings to the board.

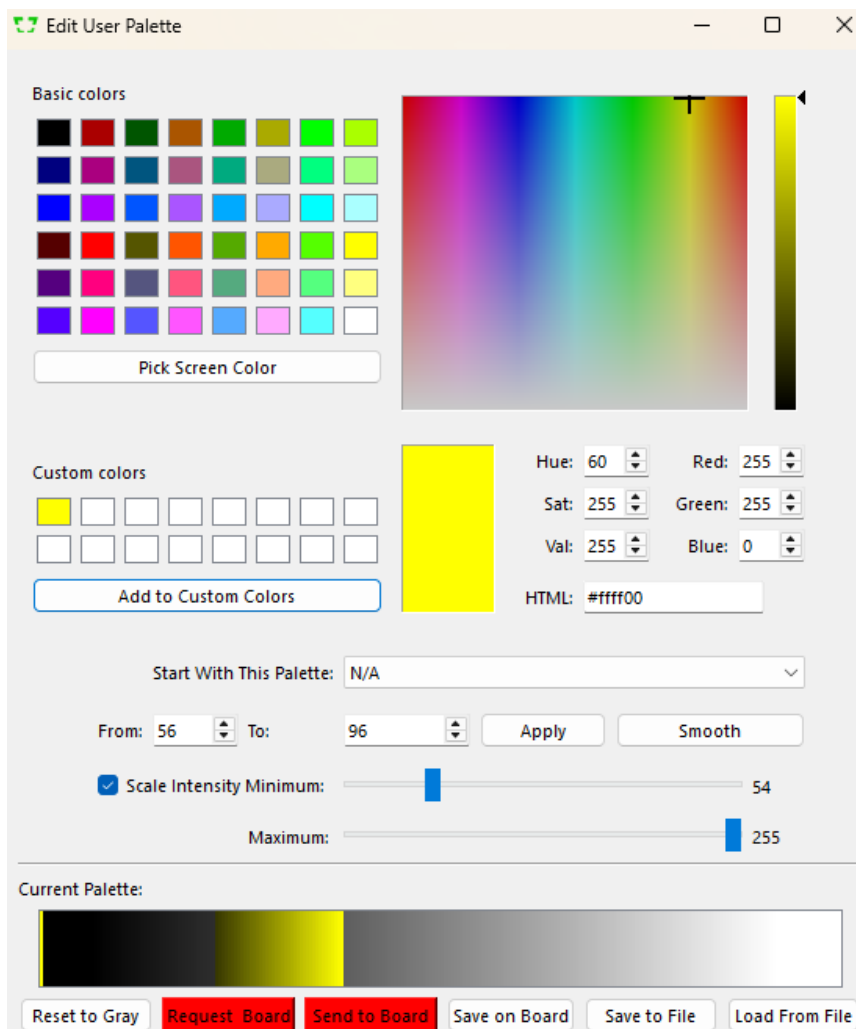


Figure 12: Setting Up a Defined Temperature Range



9 Autogain Update Rate Adjustment

If the spread and center values were calculated and applied on every frame, the image levels would change too quickly, and the video quality would degrade. Instead, the minimum and maximum values of the display range are calculated and then passed through a smoothing filter. The time constant of this filter is determined by *agRate8*, which is set using [SetDigitalCameraParameters \(0x70\)](#). The default value is 32, which corresponds to a smoothing time constant of ~1 second. These smoothed values are then used to calculate the spread and center for the current auto gain calculation

 *To freeze the auto gain, or preset a specified max and min count, overwrite the filter output with a constant max and min value from the Capture tab in Panel Plus.*

10 16-Bit Grayscale Snapshot

It is possible to capture snapshots of the raw 16-bit camera data to either the microSD card, FTP server, or to the 4000-OEM/4100-OEM USB device. This will enable conversion of each pixel element of the snapshot into full resolution temperature data offline. See the [EAN-File-Recording](#) for more information.

11 Questions and Additional Support

For questions and additional support, please contact [Sightline Support](#). Additional support documentation and Engineering Application Notes (EANs) can be found on the [Product Support Documentation](#) page in the [Sightline Knowledge Base](#).